

A chaotic image encryption algorithm based on perceptron model

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Received: 16 March 2010 / Accepted: 28 May 2010 / Published online: 22 June 2010
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Abstract Based on the high-dimension Lorenz chaotic system and perceptron model within a neural network, a chaotic image encryption system with a perceptron model is proposed. This paper describes the algorithm flow in detail, and analyses the cryptographic security. The experimental results show that this algorithm has high security, and strong resistance to the existing attack methods.

Keywords High-dimension chaotic system · Chaotic encryption · Neuron · Perceptron model

1 Introduction

Chaos has become one of the most important research fields in the past decade years. Almost all the scholars are trying to apply it in the secure communication and cryptography field with the chaotic control or synchronization [1–3]. In the information subject, chaos can be divided into two categories: one is the chaotic secure communication system with chaotic synchronization as the technical core [4–6], the other one is to construct a new digital chaotic encryption algorithm using

a chaotic system based on the computer finite precision technology [7–11]. To the digital chaotic encryption algorithm, there are two typical examples: one is proposed by Baptista in 1998 [7] and the other one is proposed by Álvarez in 1999 [8]. Since then, there are always new chaotic encryption algorithms and various improved methods have been proposed. Because of the continuous development of chaotic cryptanalysis, the existing chaotic encryption algorithms, most of them have potential security hazard more or less [12–14].

Anyway, it is necessary to propose and design better ones in order to avoid the threats and resist attacks. Therefore, this paper introduces the perceptron conception of a neural network to a chaotic encryption system, and proposes a new chaotic encryption algorithm. The experimental results show that this algorithm has a higher key sensitivity and better robustness.

2 System description

As we know, using the low-dimension chaotic system to design one encryption algorithm is very easy, but some defects of the low-dimension chaotic system, like the simple dynamic characteristic, simple cipher code, and less determine sequence parameters, may cause serious security problems. Compared with the low-dimension ones, the high-dimension chaotic system has a more complex structure, more system variables, and parameters. All of these characters assure it can be used as a cipher key of cryptosystem. Then the

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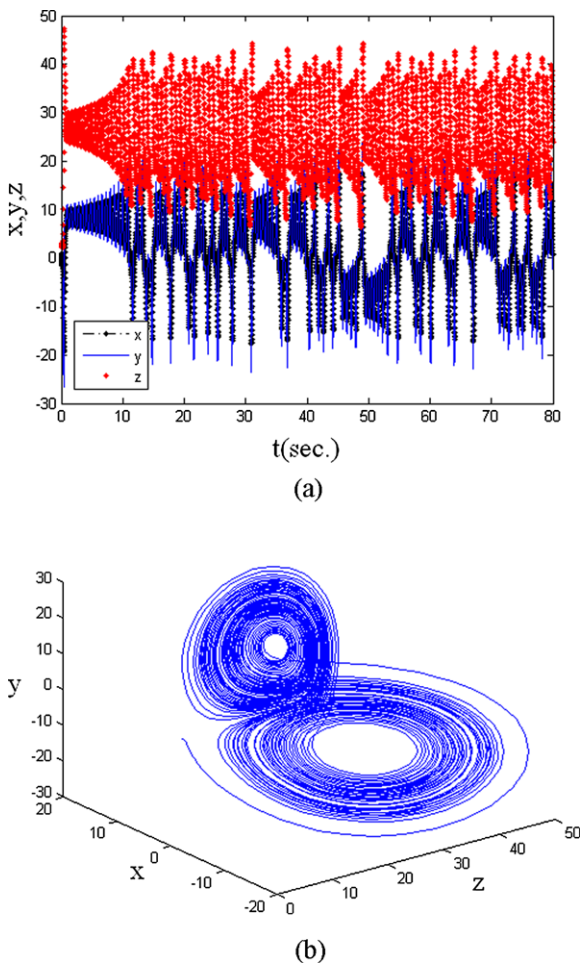


Fig. 1 Chaotic dynamic behavior of Lorenz system: (a) Chaotic track of Lorenz system, (b) chaotic attractor of Lorenz

cryptosystem's key space will be larger, and the system variables' time sequence will be more erratic and unpredictable than using the low-dimension chaotic system. So it is better to choose a high-dimension chaotic system to design the cryptosystem. This paper chooses the Lorenz system

$$\begin{cases} \dot{x} = a(y - x) \\ \dot{y} = cx - xz - y \\ \dot{z} = xy - bz \end{cases} \quad (1)$$

as the encrypttext. The Lorenz system has a complex inner structure. When the system parameters vary, the system will show different period's chaotic dynamic behavior. Figure 1 shows the dynamic behavior of the Lorenz system when parameters are $a = 10$, $b = 8/3$, $c = 28$ [15, 16].

3 Perceptron model

For each neuron, introduce a set of input signals, and each signal corresponds to one weight. All input weights determine the neuron's output [17]. When the neuron has accessed network input, there should be an appropriate output. Each neuron should have a threshold value. This paper uses the following threshold function:

$$f(net) = \begin{cases} 1, & \text{if } net > \theta \\ 0, & \text{if } net \leq \theta \end{cases} \quad (2)$$

Single layer perceptron is a simple single-layer neural network. There are no information flows among the internal neurons, and the weights of the neuron can be changed as the network changes. There are n neurons and n inputs $x_1, x_2, \dots, x_n$ to compose the network of perceptron model, w_{ij} denotes the weight of the i th input about the j th neuron, θ_j is threshold value of the j th neuron, and the output of neuron is

$$y_j = \begin{cases} 1, & \text{if } \sum_{i=1}^n x_i w_{ij} \geq \theta_j \\ 0, & \text{else} \end{cases} \quad (3)$$

4 Algorithm

In this paper, in order to adjust each neuron's weight of the single layer perceptron dynamically and finally output the ciphertext, a high-dimensional chaotic map is used to generate pseudo-random sequence. The specific algorithm is as follows:

Step 1: Let Lorenz system (1) iterate 3001 times. Throw away the numbers of previous $m = 3000$ times and keep the 3001th value as initial value (X, Y, Z) of Lorenz system.

Step 2: Do 8 iterations to get (X_k, Y_k, Z_k) , $k \in [1, 8]$. According to the following (4) and (5), normalize X_k, Y_k to $[0, 1]$

$$x_k = (X_k - X_{\min}) / (X_{\max} - X_{\min}) \quad (4)$$

$$y_k = (Y_k - Y_{\min}) / (Y_{\max} - Y_{\min}) \quad (5)$$

Then calculate the corresponding values x_k and y_k . Here, $X_{\max} = \max\{X_k\}$, $X_{\min} = \min\{X_k\}$, $Y_{\max} = \max\{Y_k\}$, $Y_{\min} = \min\{Y_k\}$ ($k = 1, 2, \dots, 8$).

In accordance with the following nonlinear transformation rules:

$$\text{key}X(k) = \begin{cases} 1, & x_k \geq 0.5 \\ 0, & x_k < 0.5 \end{cases} \quad (6)$$

$$\text{key}Y(k) = \begin{cases} 1, & y_k \geq 0.5 \\ 0, & y_k < 0.5 \end{cases} \quad (7)$$

the results $\text{key}X(k)$, $\text{key}Y(k)$, $k \in [1, 8]$ will become perceptron weight parameters. Select 8 bits in Z_8 randomly to create m , and then use the following equations to enlarge the periodicity of the Lorenz system. In order to enlarge the periodicity of the high-dimension Lorenz chaotic system, (8) and (9) are used to do the next chaotic iteration to get new X and Y value, and Z can be set to the previous value Z_8

$$w_i = \sum_{k=0}^7 \text{key}X(k)2^k, \quad w_j = \sum_{k=0}^7 \text{key}Y(k)2^k \quad (8)$$

$$\begin{cases} X = ((w_j \oplus m)x_8/256)(X_{\max} - X_{\min}) + X_{\min} \\ Y = ((w_i \oplus m)y_8/256)(Y_{\max} - Y_{\min}) + Y_{\min} \end{cases} \quad (9)$$

Step 3: Utilize the perceptron weight parameters $\text{key}X(k)$, $\text{key}Y(k)$, $k \in [1, 8]$ to do the following nonlinear transformation:

$$w_{1,k} = \begin{cases} 1, & \text{if } \text{key}X[k] = 1 \\ -1, & \text{else} \end{cases} \quad (10)$$

$$w_{2,k} = \begin{cases} 1, & \text{if } \text{key}Y[k] = 1 \\ -1, & \text{else} \end{cases} \quad (11)$$

The result is each neuron's weights of perceptron, and the weights can be written as $w_{1,k}$ and $w_{2,k}$

$$c_k = \begin{cases} -0.5, & w_{1,k} = 1 \\ 0.5, & w_{1,k} = -1 \end{cases} \quad (12)$$

$$w'_{1,k} = \begin{cases} 1, & w_{1,k} = 1 \\ 0, & w_{1,k} = -1 \end{cases} \quad (13)$$

$$w'_{2,k} = \begin{cases} 1, & w_{2,k} = 1 \\ 0, & w_{2,k} = -1 \end{cases} \quad (14)$$

$$\theta_k = w'_{1,k} \oplus w'_{2,k} \quad (15)$$

Then the other input c_k and threshold value θ_k ($k \in [1, 8]$) can be obtained.

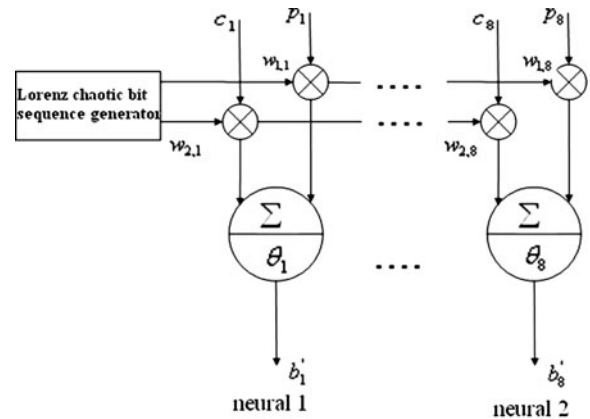


Fig. 2 The encryption algorithm flow

Step 4: Using stream cipher strategy to encrypt image. Take one pixel p from the image. b_k ($k \in [1, 8]$) denotes the k th bit of pixel p 's value b . After encryption, the ciphertext's pixel value is b' , and b'_k ($k \in [1, 8]$) denotes the k th bit of pixel value b' . The ciphertext is

$$b'_k = \begin{cases} f(w_{1,k}b_k + w_{2,k}c_k - \theta_k), & w_{1,k} = 1 \\ f(w_{1,k}b_k - w_{2,k}c_k + \theta_k), & w_{1,k} = -1 \end{cases} \quad (16)$$

where

$$f(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

Step 5: Repeat step 2–step 4, until the whole image is encrypted. Figure 2 shows the encryption flow.

5 Security analysis

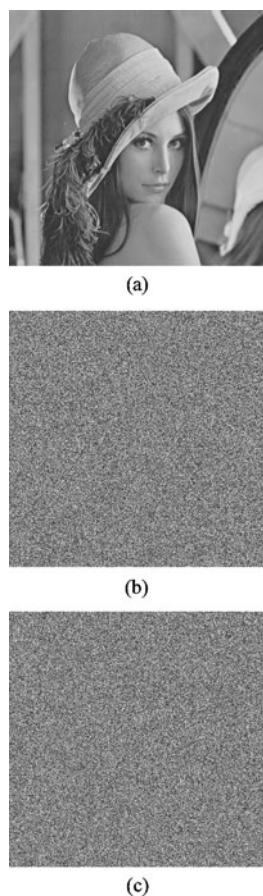
Experiment environment: CPU: PentiumIV 1.4 G, Memory: 512 M, Operating system: Windows XP, Program language: C++.

Choose the system parameters $a = 10$, $b = 8/3$, $c = 28$, and initial value $X = 0$, $Y = 0$, $Z = 10^{-10}$. Figure 3 shows the experimental results after using this algorithm to encrypt the image of Lena.

5.1 Histogram analysis

Figures 4(a) and 4(b) are the histograms of Figs. 3(a) and 3(b), respectively. Compare Fig. 4(a) with Fig. 4(b); the latter histogram is uniformly distributed

Fig. 3 The encryption results of Lena: (a) Original image, (b) initial value $z = 10^{-10}$, (c) initial value $z = 2 \times 10^{-10}$



after using the proposed encryption algorithm and it masks Fig. 4(a)'s distribution rule, increasing the difficulty of deciphering.

5.2 The correlation of adjacent pixels

As we know, the correlation of adjacent pixels of original image is so big that it can be broken easily by statistical attack. 1000 pixel-pairs are selected from Fig. 3(a) and Fig. 3(b) randomly to test the horizontal and vertical correlation. Figure 5 shows the experimental results about the correlation of adjacent pixels before and after encryption. Table 1 lists the correlation coefficient of vertical direction.

As Fig. 5 and Table 1 show, the correlation of adjacent pixels of cipher-image is much lower than that of plain-image.

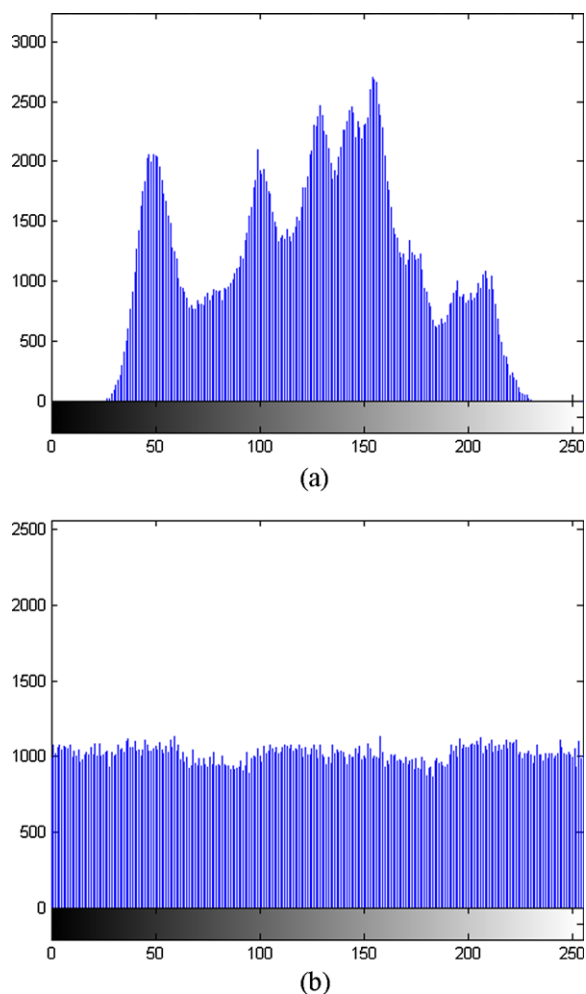


Fig. 4 Histograms of Lena: (a) Before encryption, (b) after encryption

5.3 Key sensitivity analysis

Add 10^{-10} to the parameter Z of system (1), and then encrypt the original image Lena. The results are shown in Fig. 3(c). Calculate Figs. 3(b) and 3(c)'s NPCR [18], and it is 0.76%. This result shows that the similarity of these two images is little. Thus, it can be concluded that this algorithm is quite sensitive to the key.

In order to check the system's key sensitivity, small changes are made to the other parameters of the keys. It can be seen from Fig. 6 that the system is extremely sensitive to the key, and even a very small difference with the right key will lead a failure of decryption.

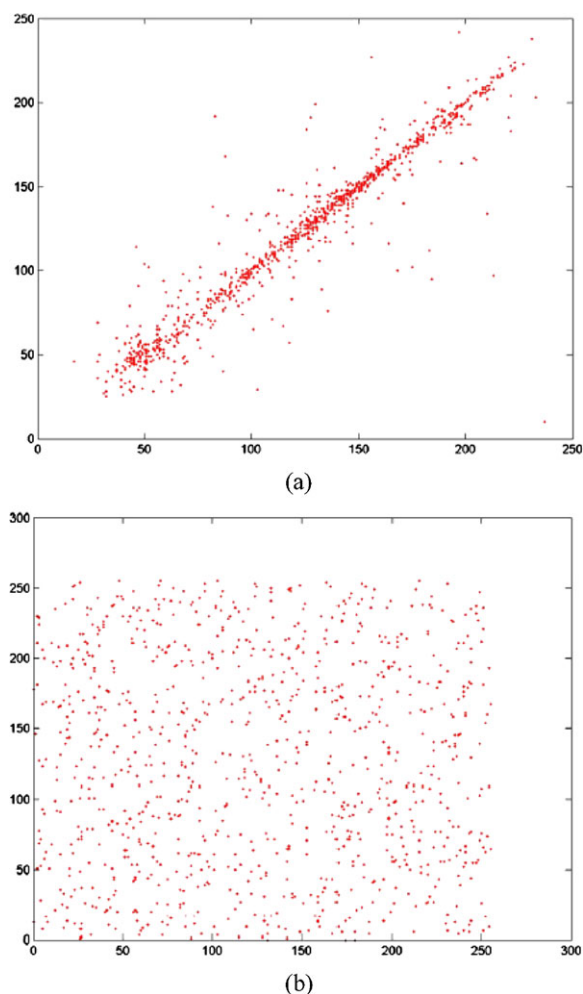


Fig. 5 Adjacent pixels' horizontal correlation before and after encryption: (a) Before encryption, (b) after encryption

Table 1 Adjacent pixels' correlation coefficients before and after the encryption of Lena

Direction	Original	After encryption
Vertical	0.9484	-0.0876
Horizontal	0.9404	0.0056

5.4 Cropped and noise attack analysis

Ciphertext may lose data or influenced by noise during transmission, so cropped and noise attack should be considered. Figure 7(b) is the decryption result when the ciphertext lose part of data. Except the dark clip part cannot be recovered; others can be seen clearly.

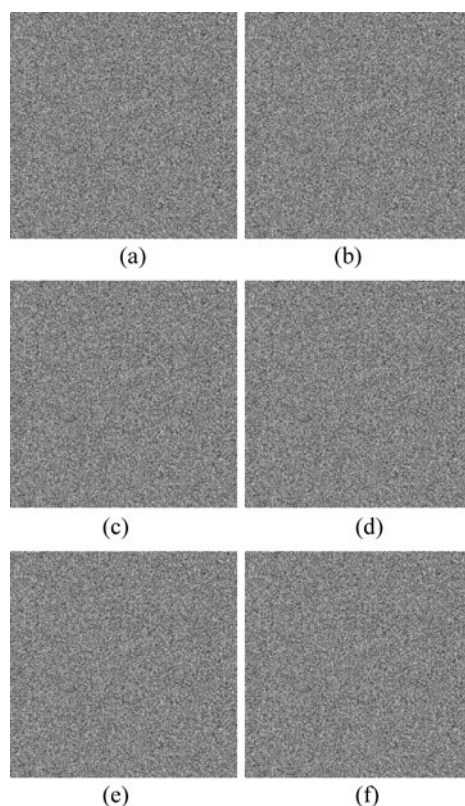
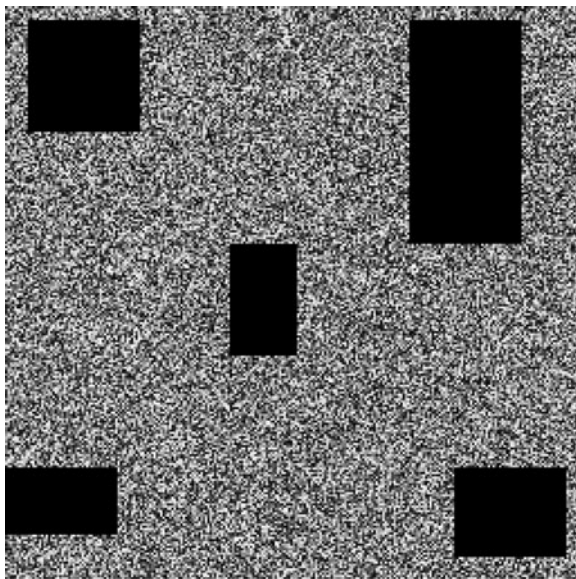


Fig. 6 Decryption results under the wrong keys: (a) $X = 0.0000001$, (b) $Y = 0.0000001$, (c) $a = 11$, (d) $b = 2$, (e) $c = 27$, (f) preiteration times $m = 2999$

Figure 8(b) is the decryption result after noise attack. From Fig. 8(b), it can be seen that the general outline of the image can be decrypted correctly. This shows that the proposed method can resist cropped and noise attack.

6 Conclusion

This paper presents a chaotic image encryption algorithm based on simple Perceptron. High-dimensional chaotic system is used to produce three sets of pseudo-random sequence, and two of them are normalized. Then a nonlinear strategy is adopted, which can generate weight of each neuron of Perceptron as well as a set of input signal. At the same time, dynamically adjust chaotic system's parameters, which greatly extend the periodicity of chaotic systems, and effectively solve the cycle state issues caused by discrete chaotic system.

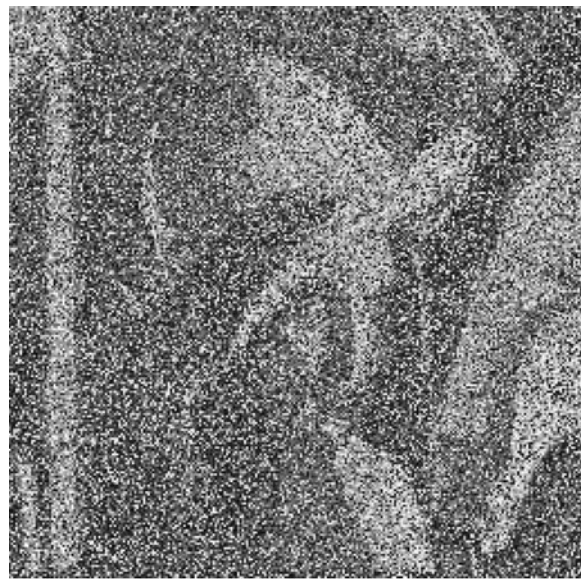


(a)



(b)

Fig. 7 Cropped attack experiment: (a) cropped attack image, (b) decryption result



(a)



(b)

Fig. 8 Noise attack experiment: (a) noise attack image, (b) decryption result

Acknowledgements This research is supported by the National Natural Science Foundation of China (No.: 60973152, 60573172), the Superior University Doctor Subject Special Scientific Research Foundation of China (No.: 20070141014), and the Natural Science Foundation of Liaoning province (No.: 20082165).

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